RESEARCH & PROTOTYPE REVIEW

Feasibility Report

Code-aware feasibility of a scoped, safe prototype and the main technical/evidence gaps.

SIH 2026 | PS SIH26250 | Alchemy Architects

SOURCE-CODE REVIEW + PUBLIC RESEARCH | 20 OCTOBER 2026

ALCHEMY ARCHITECTS

Executive verdict

Feasible as a scoped student demonstrator; not yet evidenced as an operational product. The source export contains a structured front end and deterministic scheduling logic. Data permissions, live integration reliability, validator completeness, model credibility, security, field validation and end-to-end runtime verification remain substantial dependencies.

1. Technical feasibility by workstream

Workstream	Feasibility	Evidence in export	Work needed
Responsive UI and navigation	High	React/TypeScript views and provided screenshots.	Smoke-test the exact hosted build on desktop/mobile and keyboard navigation.
Generic planning heuristic	Medium-high	Custom TypeScript scheduler, candidate scoring and greedy baseline.	Clarify objective, enforce all hard constraints and compare like-for-like with baseline.
Independent validation	Medium	Separate validator function with several check categories.	Add deadline/earliest start/duration/task identity/accounting tests and independent adversarial cases.
Public civil weather	Medium	Server path calls AviationWeather.gov API, with fallback behaviour.	Verify endpoint response, terms, source times, fields, timeout and failure signals.
IMD data	Low-medium	UI/service expects a local API; source includes static baseline records.	Secure access/terms and distinguish live records from seeded fallback; do not fake live status.
C-MAPSS predictive maintenance	Low at present	Static screen, seed records and hard-coded metrics.	Implement dataset parsing, leakage-safe training, inference and metric recomputation.
Gemini assistant	Medium as optional component	Server-side request path and deterministic fallback in source.	Verify model ID support, key handling, response parsing, grounding and operational scope filters.
Firebase / access control	Low-medium	Firebase config and Firestore rules were included.	Use restricted client key, owner checks, default deny, rules tests and secure credentials.
Field/user validation	Unknown	No visit notes or signed interviews supplied.	Obtain permission and document real stakeholder observations.
Production/defence deployment	Not established	Student app only; no certification evidence.	Out of scope for current prototype; separate security, safety and authorization path needed.

2. Build and integration risks

- The hosted URL could not be fetched by the audit browser path. Source ZIP and screenshots were inspected instead; live end-to-end state is therefore unknown.
- The original package includes a Google AI Studio/Bun dependency set that differs from the curated GitHub branch. Treat the tested safe-demo CI as separate evidence from the original export.
- A static fallback can preserve display availability but must not label a cached or seeded record LIVE or VERIFIED without a documented source observation.
- The API endpoint for the predictive lab returns metric metadata and the source data include fixed examples; the actual regression pipeline remains to be built.
- Original Firestore rules included broad reads over collections described as tactical discrepancies and operational status. Use restrictive rules and do not store sensitive data.
- Source includes operationally themed/weapon-oriented screens; the public SIH demo should be restricted to generic, synthetic, human-reviewed planning.

3. Minimum viable implementation plan

Sprint	Deliverables	Acceptance gate
Sprint 1 - Core correctness	Canonical schemas, import checks, hard constraint list, deterministic solver, complete independent validator.	Feasible/infeasible cases; all invalid injected candidates rejected; tests pass.
Sprint 2 - Data provenance	One public civil-weather connector, timestamp/status display, no fabricated fallback, clear licence/source notes.	Saved raw response, schema validation, stale/failure path tests.
Sprint 3 - Dynamic comparison	Baseline/candidate metrics from one evaluator, scenario replay, versioned input/output evidence.	Same inputs produce repeatable comparison; no metric constants pretending to be results.
Sprint 4 - ML experiment	Actual NASA data, preprocessing, unit-wise split, baseline and candidate model, metric script.	Independent reviewer can reproduce RMSE/MAE and uncertainty methodology.
Sprint 5 - User research	Authorized office/user consultation, field notes and usability test.	Dated notes and permission; at least one workflow improvement derived from feedback.
Sprint 6 - Release hygiene	Secret scan, data licence review, dependency lock, install/build/test CI, README with implemented/planned status.	Clean clone build, all tests pass, safe public scope approved by team.

4. Feasibility summary

Question	Answer
Can a useful student prototype be demonstrated?	Yes, if limited to generic synthetic resource scheduling, independent checks, scenario replay and evidence export.
Can current UI prove an accurate trained predictor?	No. The displayed RUL metrics need a reproducible training/evaluation pipeline.
Can the UI claim live connected data?	Only for the exact endpoint and run whose response, timestamps and provenance are verified.
Can it claim to reflect a defence office workflow?	No, not without authorized stakeholder evidence; public civil aviation offices provide analogues only.
Can it be represented as production/operational ready?	No. The current evidence is insufficient for safety-critical or defence operational use.

References and source notes

[1] AviationWeather.gov Data API
https://aviationweather.gov/data/api/
Official public civil aviation weather data API documentation; connector status and terms must be verified before live claims.

[2] NASA Prognostics Center of Excellence Dataset Repository
https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/
Official NASA repository lists turbofan engine degradation simulation data and the C-MAPSS benchmark. It is not military telemetry.

[3] Airports Authority of India - Madurai Airport contacts
https://www.aai.aero/en/airports/contact-us/madurai
Official public contact details for the Airport Director office and terminal manager; proposed consultation target only, not evidence of a visit.

[4] India Meteorological Department - RMC Chennai
https://mausam.imd.gov.in/chennai/
Official regional weather observations, nowcasts, forecasts and public contact details; proposed consultation target only.

[5] AAI Civil Aviation Training College - Contact
https://catc.aai.aero/en/contact
Official address and contact information for the Civil Aviation Training College, Prayagraj.

[6] AAI Civil Aviation Training College - Training Infrastructure
https://catc.aai.aero/en/training-infrastructure
Official description of air traffic management and communication/navigation/surveillance training infrastructure.